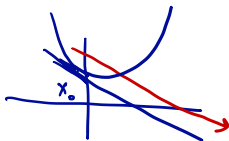


Gradient descent

minimize a linear (1st order Taylor) approximation

$$\boxed{\begin{array}{l} \min: f(x) \\ \text{st: } x \in \mathcal{X} \end{array}}$$

$$f(x) \approx \text{linear function} = f(x_0) + \underbrace{\langle \nabla f(x_0), x - x_0 \rangle}_{\text{linear approximation}}$$



Idea: start x_0

at iteration t

$$x_{t+1} = \underset{x}{\operatorname{argmin}}: f(x_t) + \langle \nabla f(x_t), x - x_t \rangle + \underbrace{\frac{1}{2\eta} \|x - x_t\|_2^2}_{\text{regularization}}$$

Take deriv (grad) set to zero:

$$0 + \nabla f(x_t) + \frac{1}{\eta} (x - x_t) = 0$$

$$\boxed{x_{t+1} = x_t - \underline{\eta} \nabla f(x_t)} \quad \eta - \text{"step size"}$$

Gradient descent - an example (1/2)

$$f(x) = 3x^2 + 4x - 2$$

— can solve this directly: $6x + 4 = 0$
 $x^* = -2/3$

Apply gradient descent.

Initialize at x_1 , take step size η .

$$\begin{aligned}\text{GD iteration: } x^+ &= x - \eta \nabla f(x) \\ &= x - \eta (6x + 4)\end{aligned}$$

$$\Rightarrow x_{t+1} = x_t - \eta (6x_t + 4)$$

$$= \underbrace{(1 - 6\eta)x_t}_{\text{}} - 4\eta$$

$$= (1 - 6\eta)[x_{t-1} - \eta(6x_{t-1} + 4)] - 4\eta$$

$$= (1 - 6\eta)^2 x_{t-1} - ((1 - 6\eta)4\eta + 4\eta)$$

$$= (1 - 6\eta)^3 x_{t-2} - ((1 - 6\eta)^2 4\eta + (1 - 6\eta)4\eta + 4\eta)$$

$\vdots$

$$= (1 - 6\eta)^t x_1 - ((1 - 6\eta)^{t-1} + (1 - 6\eta)^{t-2} + \dots + 1) 4\eta$$

$$= (1 - 6\eta)^t \cdot x_1 - \frac{1 - (1 - 6\eta)^t}{1 - (1 - 6\eta)} 4\eta$$

$$\begin{aligned}\alpha^{t-1} + \alpha^{t-2} + \dots + \alpha + 1 \\ = \frac{1 - \alpha^t}{1 - \alpha}\end{aligned}$$

Example continued:

$$x_{t+1} = (1-6\eta)^t \cdot x_1 - \frac{1 - (1-6\eta)^t}{1 - (1-6\eta)} \cdot 4\eta$$

$$\text{Need: } |1-6\eta| < 1. \Rightarrow (1-6\eta)^t \rightarrow 0 \text{ as } t \rightarrow \infty$$

$$= (1-6\eta)^t \cdot x_1 + \frac{(1-6\eta)^t}{6\eta} \cdot 4\eta - \underbrace{\frac{1}{6\eta} \cdot 4\eta}$$

$$= \underbrace{(1-6\eta)^t \left[x_1 + \frac{2}{3} \right]} - \frac{2}{3} \rightarrow -\frac{2}{3} \text{ as } t \text{ grows.}$$

$$x_t \rightarrow -\frac{2}{3} \quad \text{"linearly"}$$

Good News: GD for

this function, converges very quickly.

error goes down with $(1-6\eta)^t$ Fast when $\eta < 1/6$.

Gradient descent - an example (2/2)

step size : small enough so that $|1 - 6\gamma| < 1$

convergence rate : linear $\varepsilon \approx e^{-ct}$

improvement at every iteration : Exercise : check that

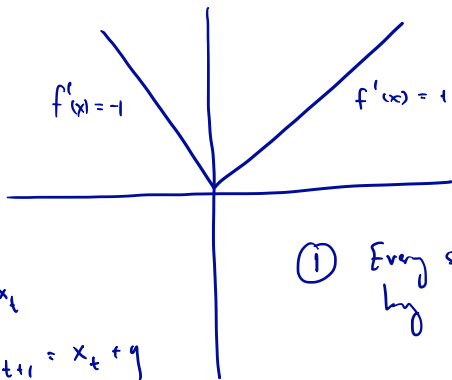
$$f(x_{t+1}) \leq f(x_t)$$

Subgradient method

$$f(x) = |x|$$

$$f'(x) = -1$$

$$f'(x) = 1$$



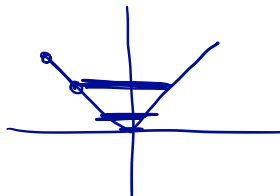
$$x_{t+1} = x_t - \eta g_{x_t}$$

$$\text{If } x_t < 0, \quad x_{t+1} = x_t + \eta$$

$$\text{If } x_t > 0, \quad x_{t+1} = x_t - \eta$$

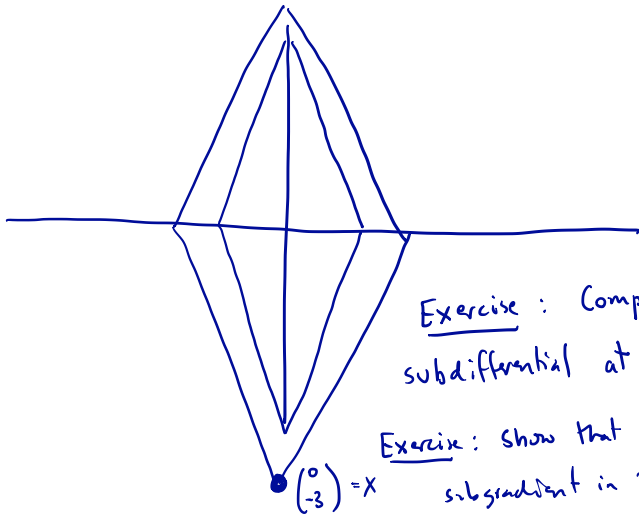
① Every step moves us closer by η when $|x_t| > \eta$.

② When $|x_t| < \eta$



Sub-Gradient method: an example

$$f(x_1, x_2) = 10 \cdot |x_1| + |x_2|$$



so that: $f(x - \eta g_x) > f(x)$



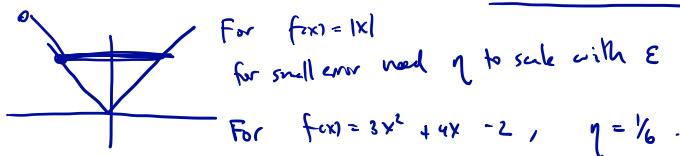
Subgradient method: a 2-d example

subgradient method is not a descent algorithm

Gradient descent vs sub gradient method

gradient descent improves at every iteration, unlike sub-gradient

gradient descent can take a "big" step size: self-tuning property



gradient descent takes bigger steps when far away

How big are the increments $x_{t+1} - x_t$

For $\boxed{f(x) = |x|}$: η

for $\boxed{f(x) = 3x^2 + 4x - 2}$

$$x_{t+1} = x_t - \eta \underbrace{(6x_t + 4)}$$